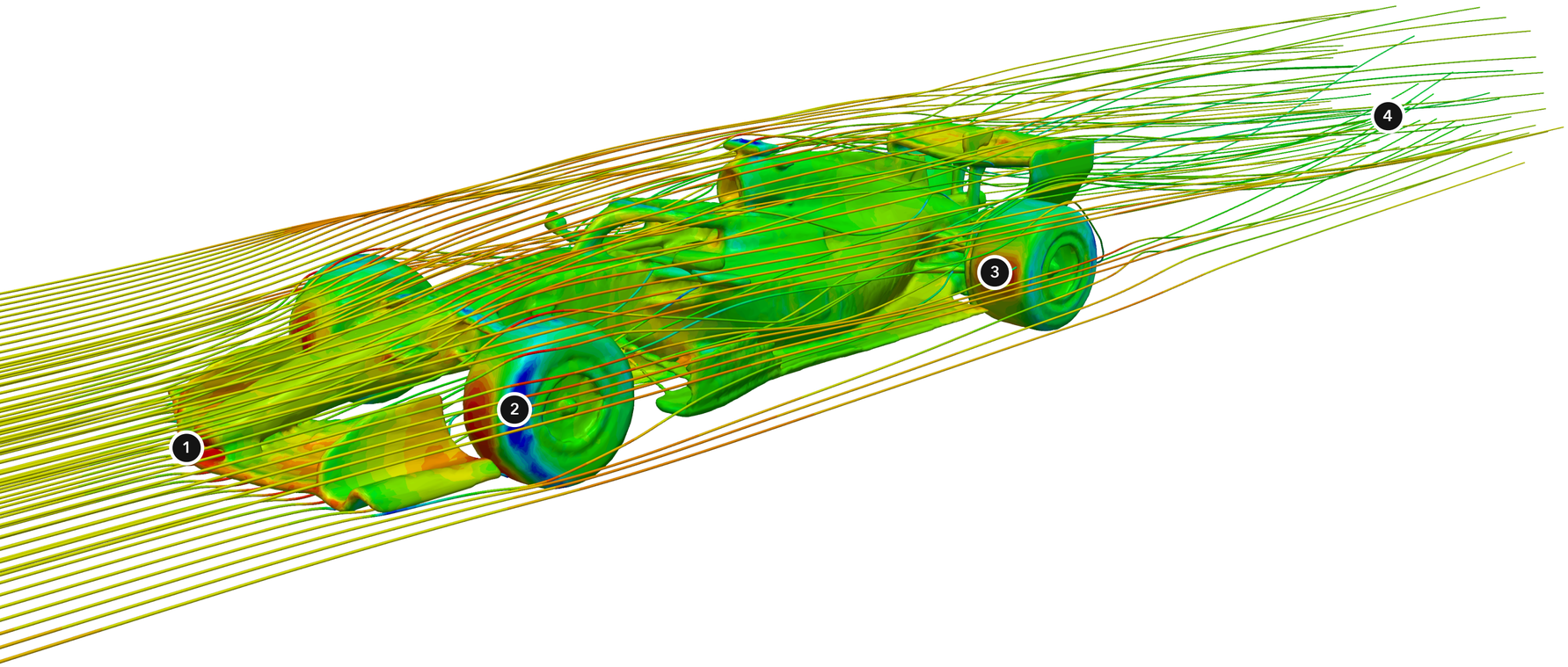
Drag is the price of speed.

At top speed, a Formula 1 car spends most of its engine power just pushing air out of the way. We solved the air around a full-size F1 car at 144 km/h, using the Navier–Stokes equations, with the wind turned through nine crosswind angles.

SDG 9

Industry, innovation and infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialisation, and foster innovation. Racing aerodynamics feeds into road cars: less drag means less power, fuel and CO_2.



$$P = \frac{1}{2}\rho C_d A V^3$$

P power · ρ air density · C_d drag coefficient · A front area · V speed

Double the speed, eight times the power.

Speed	Drag	Power to push the air
100 km/h	602 N	17 kW
200 km/h	2,407 N	134 kW
300 km/h	5,415 N	451 kW

From our result for this car, C_dA = 1.27 m².

+3.6 %

Cut drag by 10 %, go 3.6 % faster on the same power.

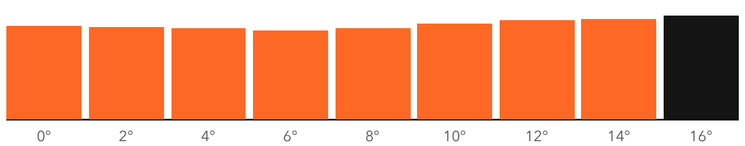
Power grows with V^3, so speed grows with the cube root: $(1 \div 0.9)^{1/3} = 1.036$.

What you're seeing

1. **Front wing.** The air hits head-on: highest pressure, in red.
2. **Front tyre.** Air races round its edge and the pressure drops, in blue.
3. **Rear tyre.** Open wheels face the wind directly, so they add drag.
4. **The wake.** Slow, swirling air behind the car holds it back.

A crosswind costs power.

+11 %



Drag at each crosswind angle: 1,247 N straight on, 1,389 N at 16° (dark). Drag power rises from 49.9 to 55.5 kW.

Surface pressure



Low: suction — High: air hits

Air speed



Still — 187 km/h

Solved with OpenFOAM, k-ω SST · 9 crosswind angles, about 390,000 cells each · 144 km/h over a moving floor

NavFlow · NPS International School STEM Exhibition